

Scene Graph-Guided SegCaptioning Transformer With Fine-Grained Alignment for Controllable Video Segmentation and Captioning

Xu Zhang¹, Jin Yuan¹, BinHong Yang, Xuan Liu¹, *Member, IEEE*, Qianjun Zhang², Yuyi Wang², Zhiyong Li², *Member, IEEE*, and Hanwang Zhang², *Senior Member, IEEE*

Abstract—Recent advancements in multimodal large models have significantly bridged the representation gap between diverse modalities, catalyzing the evolution of video multimodal interpretation, which enhances users’ understanding of video content by generating correlated modalities. However, most existing video multimodal interpretation methods primarily concentrate on global comprehension with limited user interaction. To address this, we propose a novel task, Controllable Video Segmentation and Captioning (SegCaptioning), which empowers users to provide specific prompts, such as a bounding box around an object of interest, to simultaneously generate correlated masks and captions that precisely embody user intent. An innovative framework, Scene Graph-guided Fine-grained SegCaptioning Transformer (SG-FSCFormer), is designed to integrate a Prompt-guided Temporal Graph Former to effectively capture and represent user intent through an adaptive prompt adaptor, ensuring that the generated content aligns well with the user’s requirements. Furthermore, our model introduces a Fine-grained Mask-linguistic Decoder to collaboratively predict high-quality caption-mask pairs using a Multi-entity Contrastive loss, while providing fine-grained alignment between each mask and its corresponding caption tokens, thereby enhancing the user’s comprehension of videos. Comprehensive experiments conducted on two benchmark datasets demonstrate that SG-FSCFormer achieves remarkable performance, effectively capturing user intent and generating precise multimodal outputs tailored to user specifications. Our code is available at <https://github.com/XuZhang1211/SG-FSCFormer>

Index Terms—Video understanding, scene analysis, multimodal interpretation, multimodal alignment.

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Xu Zhang, Jin Yuan, BinHong Yang, and Xuan Liu are with the College of Computer Science and Electronic Engineering, Hunan University, Changsha 410082, China (e-mail: xuan_liu@hnu.edu.cn).

Qianjun Zhang is with the School of Computing and Artificial Intelligence, Southwest Jiaotong University, Chengdu 611756, China (e-mail: zqjblue@foxmail.com).

Yuyi Wang is with CRRC Zhuzhou Institute Company Ltd., Zhuzhou, Hunan 412001, China.

Zhiyong Li is with the School of Robotics and the National Engineering Research Center of Robot Visual Perception and Control Technology, Hunan University, Changsha 410082, China.

Hanwang Zhang is with Nanyang Technological University, Singapore 639798.

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I. INTRODUCTION

VIDEO understanding [1], [2], which entails the precise extraction of semantic information from temporal visual data, holds immense potential for applications in autonomous driving [3], [4], human-computer interaction [5], [6], robots [7], [8], and intelligent surveillance [9], [10]. Current video understanding tasks encompass video segmentation [11], [12], [13], [14], [15], action detection [16], [17], and captioning [18], [19], [20], among others. In these tasks, videos are transformed into semantic representations, such as masks, tags, and captions.

With the rapid advancement of large model architectures, particularly in their ability to represent modalities and perform cross-modal transformations, video understanding has transitioned from traditional single-task, unimodal outputs to multitask, multimodal outputs. These multimodal representations exhibit intricate interdependencies between modalities, effectively harnessed by multimodal large models to provide more comprehensive and nuanced feedback, thereby enhancing user comprehension. For instance, as depicted in Fig. 1 (a), [21] associates each video instance with both a corresponding caption and mask, delivering a dual visual-textual interpretation of the video content. While these innovations have undoubtedly expanded the informational breadth available to users for video comprehension, they still lack direct user interaction. Consequently, the generated outputs often contain superfluous information that may not fully align with specific user requirements, thereby limiting their utility in precise, context-driven applications.

To address this gap, this paper introduces a novel task, “Controllable Video Segmentation and Captioning” (SegCaptioning), which empowers users to provide a visual prompt—such as a bounding box delineating an object of interest—to concurrently generate a caption and several corresponding masks across temporal frames, as depicted in Fig. 1 (b). Through this prompt-driven paradigm, the task challenges the model to autonomously identify all potential objects of interest and accurately model their spatio-temporal interrelationships, ensuring alignment with user-specific intent. Specifically, bounding boxes are adopted as prompts because they provide a favorable trade-off between user interaction effort and spatial localization precision, whereas points only offer coarse spatial cues, and text prompts typically increase

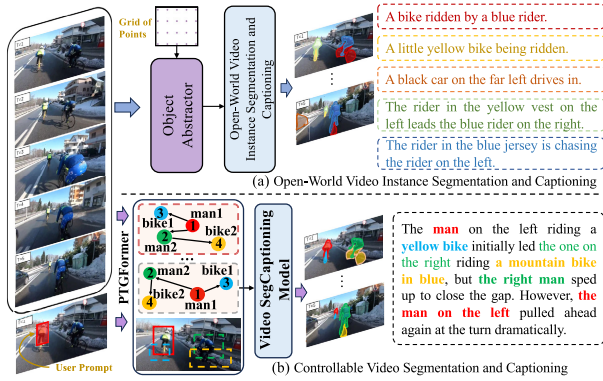


Fig. 1. An example to illustrate the difference between the segmentation and captioning method and our approach, which allows users to provide a bounding box to generate multimodal outputs that are tailored to the user's intent.

the user's annotation burden. Nonetheless, our framework can be readily extended to alternative prompts, such as points or text, through unified encoding or semantic matching strategies, as explored in Referring Expression Comprehension (REC) [22] and Referring Video Object Segmentation (RVOS) [23]. Beyond these conventional tasks, our setting emphasizes the joint generation of a user-intended narrative and instance-level masks, requiring precise correspondence between each generated mask and its associated caption token, which poses significant challenges for robust cross-modal fine-grained alignment.

To this end, this paper introduces a pioneering, Scene Graph-guided Fine-grained SegCaptioning Transformer (SG-FSCFormer), which leverages structured scene graphs to translate user intent from a basic prompt into an intent-specific subgraph that is used to guide the generation of correlated captions and masks for precise multimodal content understanding tailored to user needs. Specifically, given a video and a user prompt, such as a bounding box around an object, we first design a Prompt-guided Temporal Graph Former (PTGFormer) to identify objects of interest and their complex relationships, represented as a prompt-related graph feature. PTGFormer incorporates a novel prompt adaptor to remove irrelevant nodes and edges from the global scene graph by thoroughly exploring the spatial and temporal correlations between each object and the prompt object, resulting in a refined representation that aligns closely with user intent. Next, we design a Fine-grained Mask-linguistic Decoder to guide the generation of captions and masks. Concretely, the decoder first employs a Graph-guided Iterative Query Former that converts graph features into language embeddings, which are subsequently processed by a frozen large language model to predict captions. For mask prediction, the revised SAM2 decoder receives both images and graph features, and then simultaneously predicts masks associated with their position information appearing in the caption. This cross-modal fine-grained correspondence is supervised by a proposed fine-grained alignment loss. Furthermore, to achieve precise alignment between masks and caption words, the Mask-linguistic decoder employs a Cross-modal Multi-entity Contrastive loss, which draws the embeddings of positive caption words and their corresponding

masks closer while distancing negative ones. Consequently, the improved cross-modal representations help the decoder to produce high-quality caption-mask pairs, enhancing user comprehension. The key innovation of SG-FSCFormer lies not in a straightforward combination of independently pre-trained models or existing algorithmic components, but in the design of a unified architecture that jointly decodes textual descriptions and segmentation masks. By performing text generation and mask prediction in a single integrated framework, SG-FSCFormer effectively alleviates the word omissions and inaccurate object segmentations that commonly arise in conventional multi-stage pipelines, leading to more consistent and reliable vision-language understanding. Extensive experiments on two annotated datasets ("LV-VIS" and "OVIS") demonstrate the effectiveness of SG-FSCFormer, achieving state-of-the-art performance surpassing existing methods. In summary, the main contributions of this work are as follows:

- We introduce a pioneering task, "Controllable Video Segmentation and Captioning", marking the inaugural controllable video multimodal interpretation. This task enables the flexible and nuanced multimodal interpretations of video content, meticulously tailored to align with specific user requirements. To support this task, we present two newly annotated datasets based on LV-VIS and OVIS, which will be publicly released to facilitate future research.¹
- We propose a unified text-mask co-generation framework, called the Scene Graph-guided Fine-grained SegCaptioning Transformer. Unlike composite frameworks that merge technologies without cross-modal interaction, our design introduces a specialized Prompt-guided Temporal Graph Former (PTGFormer) that filters out irrelevant nodes from a global scene graph to capture exactly what the user wants to interpret.
- We devise a Mask-linguistic Decoder to collaboratively generate accurate, coherent multimodal outputs using a Multi-entity Contrastive loss. The outcomes achieve instance-level cross-modal alignment, significantly enhancing the user's comprehension of video content.

II. RELATED WORK

A. Controllable Video Captioning

Recent advancements in vision-language models (VLMs) have significantly enhanced models' understanding of visual content, leading to substantial improvements in video captioning [1], [2], [24], which aims to describe global video content in natural language [25], [26]. However, this global captioning paradigm fails to satisfy users' personalized requirements, motivating the development of controllable video captioning [18], [27] to specify visual elements guided by user prompts. For example, [28] uses an exemplar sentence to directly control syntax and style during captioning, while [29] introduces a video caption editing (VCE) task that automatically revises

¹<https://github.com/XuZhang1211/SG-FSCFormer>

an existing video description based on user requests. Beyond text prompts, [30] enables users to specify visual bounding boxes for subject-focused descriptions. To achieve fine-grained control over caption length or topic, [27] employs multi-hot vector representations to precisely regulate caption length, while [19] proposes a topic-guided model to generate topic-oriented descriptions. In contrast, our method jointly generates captions and masks with precise alignment, delivering more meaningful and user-tailored video understanding.

B. Referring Video Segmentation

Referring video object segmentation (R-VOS) is a specialized subfield of video segmentation that seeks to localize and segment target objects across all video frames given a user-provided prompt [31], [32], [33]. Classical video segmentation focuses on segmenting, tracking, and classifying objects drawn from a fixed set of training categories [12], [34], [35]. Recent efforts have moved beyond this closed-set assumption toward open-vocabulary video instance segmentation (OV-VIS) [36], [37], [38], [39], which leverages the vision-language alignment capabilities of vision-language models (VLMs) [40]. For example, OV2Seg [37], trained on image datasets, can be directly applied to videos at test time to recognize novel categories, while OVFormer [38] improves generalization via a lightweight module that aligns query embeddings with CLIP image embeddings, narrowing the domain gap. Although OV-VIS alleviates the limitations of predefined taxonomies, it still does not guarantee segmentation that is tailored to diverse user prompts and intent. R-VOS [41], [42], [43], [44] addresses this gap by conditioning segmentation on natural-language (or other) prompts; however, effectively modeling object dynamics and long-range temporal correlations remains challenging. To address this, [45] introduces a temporal collection-distribution mechanism that facilitates interactions between the reference token and object queries, while [31] suggests decoupling video-level referring expressions into static and motion components to enhance temporal comprehension. To adapt to novel scenes, [46] proposes an effective few-shot R-VOS model to enable rapid semantic learning and adaptation across diverse scenarios. [13] explores the use of a pretrained text-to-video (T2V) diffusion model, incorporating specially designed components for R-VOS. Besides text prompts, SAM2 [47] first introduces an interactive video segmentation method built on SAM [48], using visual prompts for high efficiency. In contrast, our method not only provides open-vocabulary labels for the predicted masks but also supports the automatic expansion of a concise user prompt into multiple semantically related objects, all organized by a semantic caption with fine-grained cross-modal alignment.

C. Video Multimodal Interpretation

Advancements in multimodal large models [40], [49], [50], [51], [52] have driven the development of video multimodal interpretation, which aims to generate correlated multimodal outputs for enhanced video understanding. These approaches can be broadly categorized into two primary paradigms. Box-caption methods, such as [26], integrate object detection, tracking, and trajectory captioning within videos by

associating instance-level bounding boxes with corresponding captions. Expanding on this idea, controllable box-caption generation incorporates user guidance to improve interpretability. For example, SMOTer [53] introduces dynamic conditional inputs to guide object trajectory generation and produce interaction-aware captions, enabling user-specified tracking granularity. Another line of research focuses on mask-caption pair generation, where methods like OW-VisCaptor [21] perform segmentation, tracking, and captioning using open-world object queries, producing instance masks paired with captions for each detected object. However, these approaches describe each identified target in isolation, overlooking the surrounding contextual information and often resulting in incomplete or fragmented descriptions. VideoGLaMM [54] generates broad and unfocused captions without precisely addressing the user's region of interest. At the same time, it relies solely on text-level referring features for mask decoding and processes each target's textual features independently within the segmentation module. This design not only amplifies inherent cross-modal discrepancies between textual and visual modalities but also aggravates ambiguities among visually similar objects, ultimately hindering accurate pixel-level segmentation.

Building on these limitations, we introduce a novel Controllable Video Segmentation and Captioning task that leverages user-specified prompts to produce correlated captions and segmentation masks. Unlike prior approaches, our framework emphasizes faithfully capturing user intent by inferring related object masks guided by a prompt box. Furthermore, it effectively bridges dynamic visual contexts with textual descriptions, enabling the collaborative generation of multimodal outputs that mitigate cumulative errors and promote complementary information exchange across modalities. Although our model utilizes established components like scene graphs and attention layers, its originality lies in their structural integration and optimization. Rather than generating descriptions and masks independently, we introduce a fine-grained alignment loss and a Multi-entity Contrastive loss to directly connect the language features of the text decoder with the spatial features of the mask decoder. This simultaneous decoding process enables different modalities to interact dynamically, establishing an end-to-end framework optimized for precise word-to-mask grounding.

III. METHOD

A. Overview

Given a video V and a user prompt P_o , such as a bounding box around an object o in the first frame, controllable video SegCaptioning extracts user-specified semantic information from the video according to P_o , which is represented as a caption sentence S paired with related masks M that cover all instances in S . We propose a novel “Scene Graph-guided Fine-grained SegCaptioning Transformer” (SG-FSCFormer), designed to simultaneously generate a caption with semantically aligned masks, which consists of three main components (see Fig. 2): a visual encoder, a Prompt-guided Temporal Graph Former (PTGFormer), and a Fine-grained Mask-linguistic Decoder (MLDecoder). The visual encoder extracts

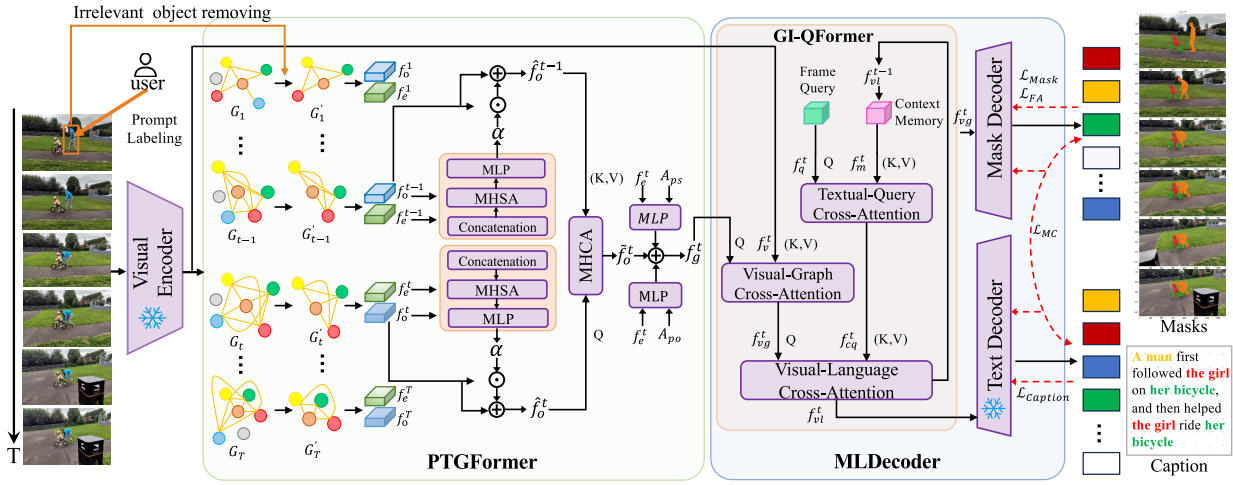


Fig. 2. Framework of our Scene Graph-guided Fine-grained SegCaptioning Transformer (SG-FSCFormer), consisting of three components: a Visual Encoder, a Prompt-guided Temporal Graph Former (PTGFormer), and a Fine-grained Mask-linguistic Decoder (MLDecoder), where the mathematical symbols are explained in the corresponding modules.

visual features for all video frames. On this basis, the PTGFormer generates temporal scene graph features to capture objects of interest and their complex relationships based on the input prompt, ensuring an accurate representation of user intent. The fine-grained mask-linguistic decoder transforms these graph features along with the encoded visual features into paired masks and captions. Notably, this collaborative decoding on both modalities is supervised by the Cross-temporal Fine-grained Alignment Loss, which aims to capture the complex fine-grained correspondence between each mask and caption words at the instance level rather than the class level, providing detailed cross-modal semantic relationships to facilitate users' understanding.

B. Prompt-Guided Temporal Graph Former

The design of the PTGFormer is motivated by the need for a structured and explicit representation of user intent. Unlike attention-based representations that lack distinct entity boundaries, scene graphs offer a powerful abstraction for modeling the complex spatio-temporal interrelationships between individual objects. By representing the video content as a graph, we can explicitly translate a user's local visual prompt into an intent-specific subgraph, enabling a more precise focus on task-relevant entities while filtering out redundant background information. Specifically, given a video $V \in \mathbb{R}^{T \times H \times W \times 3}$ with height H , width W , and T frames, our PTGFormer generates temporal scene graph features to capture the visual content of interest to users across the frames, as well as the complex semantic context embedded within them. For each frame F_t , the PTGFormer first generates a global scene graph $\{G_t\}_{t=1}^T = \{(N_t, E_t)\}_{t=1}^T$ following [55], where N_t denotes the set of object nodes $\{o_i^t\}_{i=1}^{L_g}$ and E_t denotes the set of directed relations $\{e_{i,j}^t\}_{i,j=1}^{L_g}$, with L_g representing the number of nodes in G_t . Each node is defined as $o_i^t = (c_i^t, b_i^t)$, where b_i^t denotes a region proposal generated by a Faster R-CNN detector and c_i^t is the corresponding object label. Each edge is defined as $e_{i,j}^t = (o_i^t, c_{i,j}^t, o_j^t)$, where $c_{i,j}^t$ denotes the predicate label predicted via an edge-context LSTM that encodes pairwise object interactions. This ensures that predicted relation labels

are conditioned on the spatial and semantic dependencies of the node pairs. To filter out redundant information in G_t that is not relevant to the user's interest, PTGFormer aims to identify an intent-specific subgraph $G'_t \subseteq G_t$, guided by the prompt P_o , to align well with the mask-caption pair (M, S) for model training. The goal of this selection is to maximize the expression below:

$$P(G'_t \stackrel{s}{=} (S, M) | G_t, P_o), \quad (1)$$

where $\stackrel{s}{=}$ indicates the semantic consistency between G'_t and (M, S) , approximated by object consistency between N_t of G'_t and (M, S) . To achieve this, we design an adaptive prompt adaptor to automatically extract a subgraph G'_t from G_t . To mitigate the impact of potential errors or spurious relationships inherent in scene graph generation, we adopt a two-stage hard-and-soft pruning strategy. In the hard filtering stage, given a prompt P_o , the adaptor first removes nodes and edges in G_t that are not connected to the prompt object, creating a coarse subgraph with node features $f_o^t \in \mathbb{R}^{L \times D}$ and edge features $f_e^t \in \mathbb{R}^{L_e \times D}$, where L, L_e represent the number of nodes and edges, and D is the feature dimension. This structural pruning retains only the nodes and edges that possess a direct path to the prompted object, effectively removing background noise and irrelevant entities. However, recognizing that spurious edges in SGG models may allow irrelevant nodes to bypass this initial filter, we introduce a second-stage soft filtering mechanism by exploring complex correlations between nodes. Concretely, for each pair of nodes (o_i, o_j) connected by edge $e_{ij} \in E'_t$, we first construct a relational triplet $\mathbf{r}_{ij}^t = [f_{o_i}^t; f_{e_{ij}}^t; f_{o_j}^t] \in \mathbb{R}^{3D}$, yielding $\mathbf{R}^t = \{\mathbf{r}_{ij}^t\}_{i,j=1}^{L_e} \in \mathbb{R}^{L \times L \times 3D}$ as the tensor of all triplet features, where $f_{o_i}^t, f_{o_j}^t$ and $f_{e_{ij}}^t$ represent the D -dimensional feature vectors of the nodes and the edge. Then, these triplet features are processed via MHSA and a mapping layer σ to produce the association score matrix $\alpha \in \mathbb{R}^{L \times L}$:

$$\mathbf{R}_{ij}^t = \begin{cases} [f_{o_i}^t; f_{e_{ij}}^t; f_{o_j}^t], & \text{if } e_{ij} \in E'_t, \\ \mathbf{0}, & \text{otherwise,} \end{cases} \quad (2)$$

$$\alpha = \sigma(\text{MHSA}(\mathbf{R}^t)), \quad (3)$$

where MHSA denotes the multi-head self-attention operation, σ represents a mapping layer implemented via an MLP, and α_{ij} reflects the association strength between nodes i and j . By deeply exploring these prompt-centric edge features alongside node features, the model effectively predicts the relevance of each entity to the user's intent. We then update the node features by incorporating this association strength, which helps emphasize the highly relevant nodes:

$$\hat{f}_o^t = (1 + \alpha) \cdot f_o^t, \quad (4)$$

where $\hat{f}_o^t \in \mathbb{R}^{L \times D}$ represents the prompt-guided reinforcement features of nodes that account for spatial correlations between each node and the prompt object. Instead of using a binary gate, our approach updates node features via a weighting operation in Eq. (4) to amplify the weight difference between intent-aligned entities and residual noise, thereby mitigating the impact of noise or irrelevant nodes.

Additionally, the adaptor further incorporates temporal correlations between neighboring frames to update the node features:

$$\tilde{f}_o^t = \text{MHCA}(\hat{f}_o^t, \hat{f}_o^{t-1}, \hat{f}_o^{t-1}), \quad (5)$$

where MHCA represents the multi-head cross-attention layers. By modeling spatio-temporal scene graphs, the enhanced feature $\tilde{f}_o^t \in \mathbb{R}^{L \times D}$ effectively captures the user's intent and the dynamic semantic information across frames, providing more accurate features for subsequent steps.

To aggregate the spatial and temporal context, we integrate the node features $\tilde{f}_o^t$ and edge features f_e^t to construct the final graph representation. We define two binary adjacency matrices, $A_{ps}, A_{po} \in \{0, 1\}^{L_e \times L}$, which capture the predicate-subject and predicate-object relationships within the subgraph G_t' , respectively. Specifically, $(A_{ps})_{ij} = 1$ if the i -th edge is a predicate originating from the j -th subject node, and $(A_{po})_{ij} = 1$ if the i -th edge points to the j -th object node; otherwise, the entries are 0. These matrices act as structural operators that multiply the edge features f_e^t and are subsequently transformed by two fully connected layers, MLP_{ps} and MLP_{po} , to ensure dimensional consistency with $\tilde{f}_o^t$. In this way, edge-level relational semantics are projected back into the node-level feature space. As a result, the final prompt-related graph feature $f_g^t \in \mathbb{R}^{L \times D}$ is calculated as:

$$f_g^t = \tilde{f}_o^t + \text{MLP}_{ps}(f_e^t A_{ps}) + \text{MLP}_{po}(f_e^t A_{po}). \quad (6)$$

This formulation ensures that f_g^t captures not only the objects relevant to the user's prompt but also their precise structural interactions within the temporal sequence. This results in a unified feature that facilitates the generation of semantically faithful and interaction-aware captions.

Unlike typical scene graph features, the generated f_g^t has the following key characteristics: First, it effectively filters out irrelevant nodes and edges based on the prompt object, thereby better capturing the user's intent. Second, the adaptor thoroughly explores the correlations between each graph node and the prompt object, as well as temporal correlations across frames. This enables the model to identify notable objects, which is crucial for making accurate caption predictions. Moreover, it enhances the model's ability to re-identify the

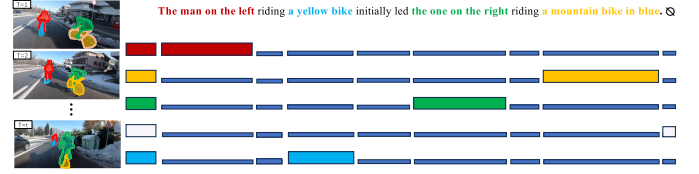


Fig. 3. Prediction by the revised decoder which is able to output the position of caption words for each mask instance.

target when it is occluded or temporarily disappears in intermediate frames.

C. Fine-Grained Mask-Linguistic Decoder

The Fine-grained Mask-linguistic Decoder is designed to generate mask regions that highlight user-specified areas of interest, along with an associated caption that semantically aligns with the predicted masks. Moreover, the decoding outcomes specify the correspondence between each mask and individual caption words, ensuring fine-grained alignment. Technically, our decoder first leverages scene graph features to guide the feature generation process, integrating both textual and visual-semantic information through a Graph-guided Iterative Query Former. Subsequently, these features are decoded to produce the corresponding caption and masks.

1) *Graph-Guided Iterative Query Former*: To effectively generate features that integrate textual and visual semantics, we employ a Graphs-guided Iterative Query Former (GI-QFormer), as Fig. 2 shows. Given a scene graph feature $f_g^t \in \mathbb{R}^{L \times D}$, an image feature $f_v^t \in \mathbb{R}^{U \times D}$ with size U , and a learnable language query $f_q^t \in \mathbb{R}^{L \times D}$ at frame t , our GI-QFormer generates language embeddings that align with user intent through three multi-head cross-attention layers: visual-graph cross-attention, textual-query cross-attention, and visual-language cross-attention. In the visual-graph cross-attention layer, f_g^t serves as the query, and f_v^t acts as both the key and value to obtain the prompt-related visual feature $f_{vg}^t \in \mathbb{R}^{L \times D}$, which accurately reflects the user's demand guided by f_g^t . In the textual-query cross-attention layer, a learnable embedding f_q^t is used as the text query, while a context memory embedding $f_m^t \in \mathbb{R}^{K \times D}$ serves as the key and value to generate a context-related language feature $f_{cq}^t \in \mathbb{R}^{L \times D}$, where K is the number of compressed temporal frames. The initial f_m^t is randomly set and updated iteratively by [25] for long-term video efficiency, based on the outputs received from GI-QFormer at each time step. Finally, in the visual-language cross-attention layer, f_{vg}^t is used as the query, and f_{cq}^t serves as the key and value to generate the prompt-related textual feature $f_{vl}^t \in \mathbb{R}^{L \times D}$. Here, f_{vg}^t acts as a user-interested visual query, dynamically retrieving relevant textual information from f_{cq}^t . Consequently, the generated f_{vl}^t captures the user's intent for caption generation. All three cross-attention layers are iteratively applied to generate the final text embedding f_{vl}^T for caption generation.

2) *Decoding*: For caption generation, we employ a frozen large language model (LLM) [56] as the text decoder. The LLM takes f_{vl}^T as input to generate a caption. The caption loss $\mathcal{L}_{\text{caption}}$ is computed as Cross-Entropy (CE) loss, as detailed in [57]. For mask generation, we employ the SAM2 decoder

[47] to generate masks by employing the original mask loss $\mathcal{L}_{\text{Mask}}$. In addition, we add a branch consisting of a two-layer MLP to predict a referring probability vector, which indicates the object location of each mask in the caption, as illustrated in Fig. 3. Specifically, the mask decoder takes the f_{vg}^t as input and outputs binary region masks $M^t \in \mathbb{R}^{N \times H \times W}$, along with a probability distribution matrix $V^t \in \mathbb{R}^{N \times L_s}$, where N denotes the number of objects and L_s represents the caption length. We introduce a fine-grained alignment loss $\mathcal{L}_{\text{FA}}$ to train the model for accurately predicting V^t :

$$\mathcal{L}_{\text{FA}}^t = \text{BCE}(V^t, Y), \quad (7)$$

where Y represents the ground-truth of the probability distribution matrix labeled in our experiments.

To collaboratively predict both segmentation and captioning results, our approach explicitly aligns cross-modal features between each mask with its corresponding caption words by using a Multi-entity Contrastive loss $\mathcal{L}_{\text{MC}}$. Given a predicted mask-caption pair (M, S) , where M consists of T sets $\{M_t\}_{t=1}^T$ for T frames, $\mathcal{L}_{\text{MC}}$ measures the correspondence between masks in M_t and words in S within each static frame F_t . Notably, there is no strict one-to-one correspondence between masks and words. For instance, the word “elephant” may correspond to multiple masks, and conversely, a single mask may relate to multiple distinct words. To account for this, we aim for the i -th mask embedding $\mathbf{m}_i^t$ to be close to the set of positive word embeddings $\mathbf{s}_i^+$, while remaining distant from unrelated word embeddings $\mathbf{s}_j$. Similarly, we expect the i -th word embedding $\mathbf{s}_i$ to be similar to the set of positive mask embeddings $\{\mathbf{m}_i^t\}^+$ and dissimilar to irrelevant embeddings $\mathbf{m}_j^t$. Consequently, $\mathcal{L}_{\text{MC}}$ is defined as:

$$\begin{aligned} \mathcal{L}_{\text{MC}}^t = & - \sum_{i=1}^{|M_t|} \frac{1}{|\{\mathbf{s}_i\}^+|} \sum_{\mathbf{s}_i \in \{\mathbf{s}_i\}^+} \log \frac{\exp(\mathbf{m}_i^{t\top} \mathbf{s}_i / \tau)}{\sum_{j \neq i} \exp(\mathbf{m}_i^{t\top} \mathbf{s}_j / \tau)} \\ & - \sum_{i=1}^{|S|} \frac{1}{|\{\mathbf{m}_i^t\}^+|} \sum_{\mathbf{m}_i^t \in \{\mathbf{m}_i^t\}^+} \log \frac{\exp(\mathbf{s}_i^\top \mathbf{m}_i^t / \tau)}{\sum_{j \neq i} \exp(\mathbf{s}_i^\top \mathbf{m}_j^t / \tau)}, \end{aligned} \quad (8)$$

where τ is a learnable parameter, $|M_t|$ and $|S|$ denote the number of mask embeddings and word embeddings at the t -th frame, respectively. Finally, the overall loss of our mask-linguistic decoder is composed of three parts by using a balancing weight λ : the captioning loss $\mathcal{L}_{\text{Caption}}$, the segmentation loss $\mathcal{L}_{\text{Mask}} + \mathcal{L}_{\text{FA}}$, and the cross-modal Multi-entity Contrastive loss $\mathcal{L}_{\text{MC}}$, which is expressed as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Caption}} + (\mathcal{L}_{\text{Mask}} + \mathcal{L}_{\text{FA}}) + \lambda \mathcal{L}_{\text{MC}}. \quad (9)$$

IV. DATASET ANNOTATION

We augment existing video segmentation datasets with textual annotations to facilitate our SegCaptioning task. Given an input video, the original dataset provides ground-truth masks and category labels for each target. We first compute the enclosing rectangle around each mask to obtain its bounding box. Annotators then review these boxes across frames and provide a fine-grained, aligned caption about the bounding target. To ensure both scalability and precision, we employ a semi-automated model-assisted human

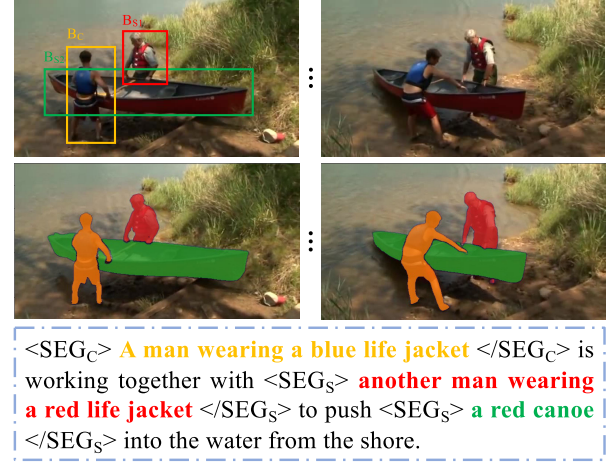


Fig. 4. An example illustrating our data annotation.

refinement pipeline. An LLM first generates initial caption candidates conditioned on the target bounding boxes, after which annotators verify correctness and refine errors, including inaccurate descriptions, box-caption misalignment, and occasional hallucinations. Within each caption, the bounding box corresponding to the subject is designated as the central box B_c , which serves as the user prompt box during training, while all other relevant boxes are treated as association boxes B_s .

For a single video, multiple prompt box-caption pairs can be created for different objects of interest. For example, in a given video, the central box surrounds “a man wearing a blue life jacket” (see Fig. 4). The required annotations are as follows: Central box $\rightarrow$ “a man wearing a blue life jacket”, Association box 1 $\rightarrow$ “another man wearing a red life jacket”, Association box 2 $\rightarrow$ “a red canoe”. To structure the caption, we use $\langle \text{SEG}_C \rangle \langle / \text{SEG}_C \rangle$ to enclose nouns corresponding to the central box and $\langle \text{SEG}_S \rangle \langle / \text{SEG}_S \rangle$ for nouns corresponding to the association box. The final annotated caption is: $\langle \text{SEG}_C \rangle$ A man wearing a blue life jacket $\langle / \text{SEG}_C \rangle$ is working together with $\langle \text{SEG}_S \rangle$ another man wearing a red life jacket $\langle / \text{SEG}_S \rangle$ to push $\langle \text{SEG}_S \rangle$ a red canoe $\langle / \text{SEG}_S \rangle$ into the water from the shore.

During the model’s training and inference, only the central box is provided as the user prompt. A temporal scene graph is then used to adaptively identify other objects associated with the target (with supervision from association box labels during training), which are subsequently used for video segmentation and caption generation. To ensure evaluation consistency and maintain a clear focus on the user-intended target, we synchronize the starting point of each training sample with the first frame in which the prompted instance appears. Specifically, if a target of interest first emerges at frame t , we treat this as the initial frame for that sample and omit the preceding frames (1 to $t - 1$) where the target is absent. This temporal alignment ensures that the user prompt is spatially grounded from the beginning of the analyzed sequence. It is important to note that this procedure does not involve the removal of any instances or videos; rather, it facilitates a fair assessment of the model’s ability to track and caption objects from their point of inception. Our model architecture remains fully capable of handling “Late Entry” scenarios, as the PTGFormer and the

subsequent decoding modules are designed to capture spatio-temporal dynamics across any arbitrary segment of the video. Our work is the pioneering study to implement the fine-grained alignment between caption tokens and masks guided by visual prompts, which inevitably introduces new annotated data. To support future research, we will publicly release the constructed dataset.

A. Annotation Efficiency and Scalability

To quantify the annotation effort, a team of eight annotators labeled 5,729 videos across the LV-VIS and OVIS datasets, producing 11,778 (box, caption) pairs. The datasets have average video durations of 4.57 s (LV-VIS) and 13.90 s (OVIS), with frames sampled at 5 FPS. The annotation process was completed within five days, enabled by focusing on key frames, leveraging existing mask annotations for precise localization, and employing a semi-automated, model-assisted human refinement pipeline. Since reviewing and refining text is significantly more efficient than manual generation, the marginal cost remains low, ensuring the feasibility of scaling this process to larger datasets or real-world applications while maintaining high semantic precision.

B. Discussing Annotation Quality, and Potential Bias

To ensure that the semi-automated pipeline maintains high scientific rigor without hindering the scalability of the dataset, we implemented several measures to safeguard annotation quality and minimize bias. First, the bounding boxes are derived directly from the precise ground-truth masks of the original datasets, ensuring exact spatial localization and eliminating potential human-induced localization bias. Second, to ensure linguistic reliability, the LLM-generated captions are subjected to a dual-stage verification: each proposal is reviewed by two independent annotators to correct semantic errors or hallucinations, followed by a final cross-check to ensure high inter-annotator agreement. This human-in-the-loop refinement process guarantees that the final descriptions are both grammatically accurate and semantically faithful. Finally, to better reflect diverse user intentions and reduce perspective bias, we allow multiple valid central boxes within each scene. Each designated central box triggers a distinct prompt-specific caption, reflecting diverse user perspectives and intent. Instead of merging these into a single consensus caption, we preserve the diversity across annotations. This approach enables the dataset to better represent nuanced shifts in user intent and provides richer supervisory signals for models aimed at relation-aware and context-sensitive understanding.

V. EXPERIMENTS

A. Datasets

We conduct our experiments on two video instance segmentation datasets, “LV-VIS” [37] and “OVIS” [11], by incorporating new prompt and caption annotations.

LV-VIS is a large vocabulary video instance segmentation dataset consisting of 4,828 real-world videos across 1,196 categories. The dataset includes annotated masks and is divided

into three subsets: 3,083 for training, 837 for validation, and 908 for testing. We have enriched this dataset by adding (box, caption) pairs. Specifically, several annotators were instructed to label one of the masked targets in each video using a bounding box in the first frame and then to describe an event centered on this target in relation to other masked objects. To provide fine-grained alignment for model training, all the masked objects were labeled with the position information to reflect their appearance in the captions. As a result, each video contains 1-3 bounding boxes with corresponding captions. This augmentation yields a total of 9,588 (box, caption) pairs, averaging approximately 1.98 (box, caption) pairs per video.

OVIS is a dataset designed for occluded video instance segmentation, which requires detecting, segmenting, and tracking instances in scenes with significant occlusions. The dataset consists of 901 videos across 25 object categories, divided into 607 training, 140 validations, and 154 testing videos. We have similarly extended this dataset to include (box, caption) pairs for each video, ensuring alignment between caption words and their visual targets. In total, we have added 2,190 (box, caption) pairs, averaging approximately 2.4 (box, caption) pairs per video.

B. Evaluation Metrics

We evaluate our model from three perspectives: the quality of caption generation, the quality of video segmentation, and the quality of the alignment between caption words and masks.

1) *Captioning Quality*: We assess the generated captions using METEOR [58], CIDEr [59], and SPICE [60]. METEOR captures fluency by incorporating synonym and paraphrase matching. CIDEr calculates the consensus between the generated text and human references using TF-IDF weighted n-grams. To evaluate structural semantic consistency, we adopt SPICE, which transforms captions into scene graphs and computes an F-score over the propositional content (objects, attributes, and relations).

2) *Video Segmentation*: We assess segmentation performance using the $\mathcal{J}\&\mathcal{F}$ mean score [61]. Specifically, given a predicted segmentation mask M and the ground-truth mask $\hat{M}$, the region similarity $\mathcal{J}$ is obtained by calculating the Intersection-over-Union (IoU): $\mathcal{J} = \frac{|M \cap \hat{M}|}{|M \cup \hat{M}|}$. To measure the contour quality, the contour accuracy $\mathcal{F}$ is calculated as the harmonic mean of contour recall R_c and precision P_c via bipartite graph matching, expressed as $\mathcal{F} = \frac{2P_c R_c}{P_c + R_c}$. This metric represents how closely the boundaries of predicted masks resemble the ground truth. Finally, the overall segmentation performance is quantified by the $\mathcal{J}\&\mathcal{F}$ mean score, defined as the arithmetic average: $\mathcal{J}\&\mathcal{F} = (\mathcal{J} + \mathcal{F})/2$.

3) *Cross-Modal Alignment*: To quantify the grounding capability between visual and linguistic modalities, we evaluate alignment at both class and instance levels. Class-level Alignment (AP): We use standard Average Precision (AP) [62] to measure the category-level classification accuracy of the masks. Instance-level Alignment (AP): This metric evaluates whether a mask M_i correctly corresponds to a specific phrase P_i in the caption. For each predicted mask, we first identify the most relevant generated phrase. We then compute the textual

TABLE I

PERFORMANCE COMPARISON WITH THE ADVANCED VIDEO CAPTIONING METHODS. WHERE THE RESULTS ON THE LV-VIS AND OVIS ARE REPRODUCED, AND THOSE ON THE YOUCook2 ARE CITED FROM THE REFERENCES

Model	LLM	LV-VIS			OVIS			YouCook2		
		METEOR	SPICE	CIDE _r	METEOR	SPICE	CIDE _r	METEOR	SPICE	CIDE _r
Vid2Seq _{CVPR23} [64]	T5-Base	16.3	24.7	107.4	18.0	33.0	106.8	9.3	7.9	47.1
SMOTer _{ECCV24} [53]	GRiT	16.9	25.0	109.6	17.8	32.8	108.1	-	-	-
MA-LMM _{CVPR24} [1]	Vicuna-7B	17.1	25.2	110.6	18.6	33.1	109.3	17.6	31.5	131.2
VideoGLaMM _{CVPR25} [54]	Phi3-Mini-3.8B	16.7	25.7	112.0	19.5	33.9	112.8	15.4	28.7	124.3
SG-FSCFormer (ours)	Vicuna-7B	19.3	26.8	112.5	21.4	35.2	113.7	20.5	33.4	139.6

TABLE II

PERFORMANCE COMPARISON BETWEEN OUR APPROACH AND THE TWO CONTROLLABLE VIDEO SEGMENTATION METHODS

Model	LV-VIS			OVIS		
	$J\&F$	J	F	$J\&F$	J	F
SAMURAI _{arXiv24} [65]	87.3	85.6	89.1	72.0	69.2	74.9
SAM2 _{ICLR25} [47]	85.6	83.3	87.9	68.7	66.0	71.3
SG-FSCFormer (ours)	87.8	85.9	89.7	74.6	72.5	76.7

similarity $S(P_i, W_{gt})$ between this phrase and the ground-truth words. A prediction is considered a correct match if $S(P_i, W_{gt}) > 0.5$, where $S(\cdot)$ represents the cosine similarity of text embeddings. We finally report the mean Average Precision (mAP) over all instances. This threshold ensures that the generated narrative is both structurally and semantically anchored to the correct visual entities.

C. Implementation Details

We use SwinB [63] as our visual encoder, and the SAM2 decoder [47] and Vicuna-7B [56] as the decoders for mask and caption generation, respectively. In the PTGFormer, the graph feature f_g^t is configured with $L = 100$ and $D = 512$, while the context memory has a memory length of $K = 10$. For the total loss function in Eq. (9), we set $\lambda = 2$ to balance the model's optimization. During training, we apply horizontal flip augmentation and resize each image to a square size of 1024×1024 . We optimize the model using the AdamW optimizer with an initial learning rate of 5.0×10^{-4} and train for 20 epochs with a batch size of 1 per GPU. All experiments are conducted on 4 A6000 GPUs.

D. Comparison With State-of-the-Art Methods

We compare SG-FSCFormer with several state-of-the-art methods in video captioning, video segmentation, and video multimodal interpretation. The results are summarized in Table I, Table II, and Table III, respectively. Note that “-” indicates that the result is not reported by the original paper or the model is unavailable.

1) *Video Captioning*: On our LV-VIS and OVIS datasets, all the captioning models are re-executed. As Table I shows, SG-FSCFormer achieves the best performance on all the metrics. Compared with approaches that generate global descriptions of entire scenes (e.g., Vid2Seq, MA-LMM, and VideoGLaMM), which tend to generate broad and unfocused captions without precisely addressing the user's region of

TABLE III

PERFORMANCE COMPARISON WITH SEVERAL ADVANCED VIDEO MULTIMODAL INTERPRETATION METHODS, WHERE $\dagger$ DENOTES THE EVALUATION USING THE INSTANCE-LEVEL AP

Model	Mask Decoder	LV-VIS			OVIS		
		AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅
GLEE _{CVPR24} [67]	MaskDINO	23.9	24.6	23.3	27.1	45.4	26.3
OVFormer _{ECCV24} [38]	Mask2Former	24.7	31.1	26.5	21.3	38.5	20.8
OW-VISCap _{NIPS24} [21]	Mask2Former	-	-	-	25.4	48.8	22.8
SG-FSCFormer (ours)	Mask2Former	25.2	31.8	27.5	27.8	49.0	27.2
VideoGLaMM _{CVPR25} [54]	SAM2	25.4	32.2	27.6	28.0	49.3	26.9
SG-FSCFormer [†] (ours)	SAM2	24.8	31.5	26.9	27.5	48.6	26.7
SG-FSCFormer (ours)	SAM2	26.0	33.6	28.3	28.7	51.4	29.1

interest, our method provides more targeted and semantically faithful outputs. Similarly, while SMOTer generates captions for individual objects, it neglects the surrounding contextual information, often leading to incomplete or fragmented descriptions. In contrast, SG-FSCFormer integrates both fine-grained object details and precise contextual cues through the user-guided visual prompt mechanism. By leveraging the proposed PTGFormer, the model proficiently translates a prompt into pertinent feature representations, thereby yielding accurate caption results that are closely aligned with user intent. To further demonstrate the generalization capability of SG-FSCFormer, we extend our evaluation to the YouCook2 dataset [66], which contains pairs of query phrases and corresponding ground-truth descriptions for each video, and compare our results with the existing official results. For each video, we select the first occurrence of a ground-truth label mentioned in the description of the initial frame as the visual prompt for our method. Under this setting, our approach consistently achieves the best performance on all the metrics, further validating its robustness and adaptability.

2) *Controllable Video Segmentation*: This experiment compares our approach with the controllable video segmentation methods. Given a video, users label a box from the target masks in the first frame for segmentation evaluation. As Table II shows, our approach demonstrates commendable segmentation performance with the highest $J\&F$ scores of 87.8 and 74.6 on the LV-VIS and OVIS datasets, respectively. We guess this improvement stems from the collaborative decoding and the multi-entity contrastive loss, which could extract robust visual features from mask prediction. In the OVIS dataset, since only the training set contains offline segmentation annotations, we reserved 100 non-overlapping training videos for validation and used the rest for training.



Fig. 5. Quantitative results by different approaches tested on the LV-VIS and OVIS datasets, where the color indicates the matching relationship between masks and words.

3) *Video Multimodal Interpretation*: As shown in Table III, our approach outperforms existing video multimodal interpretation methods, achieving AP scores of 26.0 and 28.7 on the LV-VIS and OVIS datasets, respectively, for class-level alignment. This performance gain stems from the integration of the proposed GI-QFormer, which effectively captures complex cross-modal correlations between textual and visual features, along with the incorporation of the fine-grained alignment loss L_{FA} and the multi-entity contrastive loss L_{MC} . These components jointly enforce consistent representations between corresponding caption-mask pairs, thereby enhancing the mask classification accuracy. In contrast, VideoGLaMM relies solely on text-level referring features for mask decoding. This limited modality interaction leads to suboptimal performance due to inherent cross-modal discrepancies and visual ambiguities between similar-looking objects, which hinder accurate pixel-level segmentation. Notably, even when adopting Mask2Former as the segmentation backbone, our model still surpasses the OVFormer and OW-VISCap baselines, both of which also utilize Mask2Former, further demonstrating the robustness and effectiveness of our framework. Moreover, our method implements the instance-level cross-modal alignment, with AP scores of 24.8 and 27.5 on the LV-VIS and OVIS datasets, respectively. These results are close to the class-level accuracy, demonstrating the effectiveness of the revised decoder. Since OW-VISCap [21] has not released its code, its results in Table III are cited from the references.

E. Qualitative Results

Fig. 5 presents the quantitative results of our method in comparison with several advanced methods. As shown in Fig. 5 (a) and (b), OVFormer and SAM2 can only generate segmentation results. Furthermore, OVFormer can only associate its output

masks with coarse-grained class labels, lacking fine-grained instance-level alignment. In Fig. 5 (c), SMOTer generates target-specific descriptions for each object in the videos, but it has limited control over video segmentation, preventing it from tailoring the results to user needs. Moreover, SMOTer's captions fail to align with specific targets in the videos, resulting in descriptions of the object's state and actions without indicating its precise location. In Fig. 5(d), VideoGLaMM produces overly simplistic captions and segmentation masks with poorly defined boundaries. When the initial captioning stage fails to correctly identify a target (e.g., the chainsaw or cabin in Fig. 5(d) left), the cascaded framework prevents the subsequent segmentation module from generating the corresponding mask. Moreover, VideoGLaMM relies solely on text-level referring features for mask decoding, and processes each target's textual features independently within the segmentation module. This design not only amplifies inherent cross-modal discrepancies between textual and visual modalities but also aggravates ambiguities among visually similar objects. Consequently, the limited interaction across modalities and targets results in suboptimal performance, hindering accurate boundary delineation (e.g., adjacent ships) and leading to target loss over time (e.g., the disappearing log). In contrast, as shown in Fig. 5(e), our method generates multiple coherent object masks along with a corresponding video caption based on a single user-provided box prompt. Furthermore, the integration of L_{FA} and L_{MC} ensures that noun entities in the generated captions are accurately aligned with their corresponding segmented objects, thereby ensuring tight cross-modal grounding. Overall, our approach delivers customized video multimodal interpretations guided by a user's visual prompt, not only describing the actions associated with the specified target but also localizing semantically related objects in the scene.

TABLE IV

PERFORMANCE COMPARISON OF PTGFORMER USING THE ADAPTOR WITH DIFFERENT COMPONENTS TESTED ON THE LV-VIS VALIDATION SET, WHERE “SPA.” AND “TEM.” REPRESENT THE USE OF SPATIAL AND TEMPORAL CORRELATIONS, RESPECTIVELY

Spa.	Tem.	SPICE	CIDEr	$J\&F$	AP
-	-	24.6	108.3	85.0	23.9
✓	-	25.7	111.2	86.5	25.4
-	✓	25.3	109.8	86.4	25.1
✓	✓	26.8	112.5	87.8	26.0

TABLE V

PERFORMANCE COMPARISON BY USING DIFFERENT COMPONENTS OF LOSSES TESTED ON THE LV-VIS VAL DATASET

L_{FA}	L_{MC}	SPICE	CIDEr	$J\&F$	AP
-	-	23.8	106.9	84.5	22.1
✓	-	25.4	109.8	86.3	25.6
-	✓	25.1	110.4	86.9	25.8
✓	✓	26.8	112.5	87.8	26.0

TABLE VI

PERFORMANCE COMPARISON BY USING L_{MC} WITH DIFFERENT WEIGHTS TESTED ON THE LV-VIS VAL DATASET

λ	0	1	2	5	10
SPICE	23.8	25.4	26.8	26.3	25.7
CIDEr	106.9	110.6	112.5	112.0	111.3
$J\&F$	84.5	85.7	87.8	86.5	86.1
AP	22.1	25.3	26.0	25.9	25.4

F. Ablation Studies

1) *Analysis of PTGFormer*: This experiment evaluates the PTGFormer incorporating an adaptive prompt adaptor to methodically filter out superfluous nodes and edges. To quantitatively assess the efficacy of this adaptor, we conduct a comprehensive analysis encompassing both spatial correlations (refer to Eq. (3) and (4)) and temporal correlations (refer to Eq. (5)), as delineated in Table IV. The experimental outcomes substantiate that the integration of spatial correlations engenders a substantial performance augmentation. By ascribing distinct weightings to graph nodes contingent upon their spatial affiliations with the prompt object, this methodology enables the derivation of more refined and semantically aligned node representations, congruent with user intent. Similarly, the incorporation of temporal correlations manifests an appreciable performance enhancement, facilitating the dynamic encoding of semantic nuances across sequential frames. The combined use of both correlations leads to substantial improvements in both segmentation and captioning tasks.

2) *Analysis of Alignment Losses*: We conduct an ablation study to evaluate the efficacy of the proposed losses, including L_{FA} and L_{MC} , as illustrated in Table V. The results substantiate that both losses significantly enhance the model’s performance. Specifically, L_{MC} enforces feature-level alignment between each mask and its corresponding caption word, establishing a robust cross-modal correspondence. Meanwhile, L_{FA} explicitly aligns the predicted masks with their corresponding instance words by modeling the mask-to-word distribution, corroborating the efficacy of the fine-grained alignment losses.

TABLE VII

ABLATION ANALYSIS OF SG-FSCFORMER ACROSS VARYING LLM BACKBONES AND MASK DECODERS

Method	SPICE	CIDEr	$J\&F$	AP
Our (Vicuna-7B&SAM2)	26.8	112.5	87.8	26.0
<i>Using Different LLMs</i>				
Our (OPT-2.7B)	25.9	111.2	87.4	25.7
<i>Using Different Mask Decoders</i>				
Our (Mask2Former)	26.3	112.0	85.9	25.2

TABLE VIII

PERFORMANCE EVALUATION OF SG-FSCFORMER REGARDING CROSS-DATASET TRANSFERABILITY AND COMPARISON WITH TWO-STAGE CASCADED ARCHITECTURES

Method	SPICE	CIDEr	$J\&F$	AP
SG-FSCFormer (ours)	26.8	112.5	87.8	26.0
<i>Cross-dataset Transferability</i>				
SG-FSCFormer (Cross-domain)	25.5	111.0	85.9	25.1
<i>Two-stage Cascaded Baseline</i>				
SG-FSCFormer (Cascaded Baseline)	25.3	110.7	86.1	24.9

Table VI further evaluates the sensitivity of λ in Eq. (9), where optimal performance is achieved at $\lambda = 2$. Increasing λ leads to a performance degradation as it diminishes the contributions of both the captioning and segmentation losses. Conversely, reducing λ weakens the cross-modal alignment, resulting in a decline in overall performance.

3) *Analysis of LLM and Mask Decoder*: To verify that our performance gains are not merely a byproduct of large-scale foundation models, we evaluate the backbone-agnostic robustness of SG-FSCFormer. By substituting our default decoders with standard, smaller-scale backbones—specifically replacing Vicuna-7B with OPT-2.7B and SAM2 with Mask2Former—we isolate the intrinsic contributions of our proposed architecture from the capacity of the underlying models.

Using Different LLMs. We evaluated the impact of replacing the Vicuna-7B backbone with OPT-2.7B [68], and report the results in Table VII (“Using Different LLMs”). When replacing Vicuna-7B with OPT-2.7B, we observe a performance drop across all metrics, indicating that stronger language modeling capabilities enhance semantic understanding in our framework. Furthermore, as shown in Table I, our model, equipped with Vicuna-7B or OPT-2.7B, consistently outperforms the MA-LMM baseline (which also uses Vicuna-7B) in the caption generation task. We attribute these gains to the synergistic effect between our scene-graph guided temporal semantic modeling and the integration of multimodal outputs.

Using Different Mask Decoders. To assess the impact of the SAM2 pretrained model in our framework, we replace it with Mask2Former, the same mask decoder used by the baseline methods. As shown in Table III and Table VII, our model equipped with Mask2Former still outperforms the baselines OVFormer and OW-VISCap, which also adopt Mask2Former. Although the performance decreases compared to that using SAM2, these results further demonstrate the effectiveness and generalizability of our framework.

4) *Cross-Dataset Transferability*: To evaluate the transferability and robustness of our framework, we conducted a

TABLE IX
PERFORMANCE ANALYSIS OF SG-FSCFORMER REGARDING
PROMPT-BOX GUIDANCE AND ALIGNMENT SUPERVISION
ON THE LV-VIS VAL DATASET

Method	SPICE	CIDEr	$J\&F$	AP
VideoGLaMM	25.7	112.0	86.3	25.4
VideoGLaMM (Alignment Supervision)	25.9	112.2	87.2	25.7
VideoGLaMM (w/ Prompt Box)	25.7	112.1	86.9	25.6
SG-FSCFormer (w/o Prompt Box)	25.4	110.8	86.5	25.3
SG-FSCFormer (ours)	26.8	112.5	87.8	26.0

cross-dataset generalization experiment. In this setting, the model was trained exclusively on the OVIS dataset and evaluated directly on the LV-VIS dataset without task-specific fine-tuning. As shown in Table VIII, our method achieves highly competitive results despite the significant domain gap between OVIS and LV-VIS. The ability to maintain high performance in a zero-shot transfer scenario confirms that our prompt-caption alignment mechanism effectively captures generalized semantic features rather than over-fitting to dataset-specific biases. These results further validate the robustness of our approach across diverse data distributions.

5) *Comparison With Two-Stage Cascaded Baseline:* To verify the necessity of our joint architecture, we implemented a two-stage cascaded baseline that performs segmentation and captioning serially. Specifically, the input prompt box is first processed by PTGFormer to identify relevant objects, after which the SAM2 decoder generates segmentation masks. These masks are then fed into the text decoder to produce captions.

As shown in Table VIII, the two-stage cascaded approach results in a noticeable performance drop across both segmentation ($J\&F$) and captioning (CIDEr) metrics. We attribute this to the lack of cross-modal synergy. In contrast, our joint framework employs fine-grained alignment losses (L_{FA} and L_{MC}) to tightly ground linguistic tokens to pixel-level masks. The cascaded baseline, however, suffers from error accumulation: mistakes in the initial segmentation stage cannot be corrected in subsequent captioning, nor can features be aligned at a fine-grained level. These results demonstrate that the proposed joint architecture is essential for achieving precise, intent-aligned multimodal interpretation.

6) *Effect of Prompt Box and Alignment Supervision:* Since our task emphasizes controllable outputs, incorporating user prompts is essential to ensure the output space strictly aligns with user intent. First, to assess the model’s intrinsic capability, we decouple the impact of the prompt box from our architectural contributions. As shown in Table IX (“w/o Prompt”), removing the prompt box leads to a performance decline as the model reverts to global semantic modeling, which is prone to generating general descriptions and masks that may miss user-intended targets. Nevertheless, even without prompt guidance, our framework remains highly competitive with VideoGLaMM. Second, to ensure a strictly fair comparison under identical input signals, we establish a “With Prompt Box” benchmark by encoding bounding box coordinates into the text query of VideoGLaMM. As shown in Table IX, even when provided with the same visual prior cue, SG-FSCFormer consistently outperforms

TABLE X
PARAMETERS AND FLOPS OF THE TRAINABLE
MODULES IN SG-FSCFORMER

Module	PTGFormer	MLDecoder
Params	10.2M	85.7M
FLOPs	1.8G	3.1G

TABLE XI
INFERENCE SPEED (FPS) COMPARISON

Method	OVFormer	SAM2	MA-LMM	SMOTer	VideoGLaMM	Ours
FPS	2.98	10.61	7.35	4.82	5.13	4.96

VideoGLaMM across all metrics. This gain is primarily attributed to the superior capability of our model to transform raw spatial coordinates into intent-aligned representations via the subgraph distillation mechanism in PTGFormer, together with the proposed alignment supervision that further facilitates fine-grained cross-modal alignment. Finally, we decoupled the impact of our specialized supervision by retraining VideoGLaMM with the proposed alignment losses. While this supervision enhances the baseline’s performance, SG-FSCFormer maintains a clear advantage, particularly in segmentation precision. This result highlights that our architectural advantages—specifically the structural pruning in PTGFormer and the joint decoding in MLDecoder—are essential for fully exploiting fine-grained alignment signals.

G. Challenging Cases Visualization

We visualize several challenging and failure cases in Fig. 6 for detailed analysis, including occlusion, blurred motion, visual ambiguity, and long-term disappearance. As shown in Fig. 6(a) and (b), when the target undergoes occasional occlusion and motion blur, our method successfully produces accurate captions and corresponding masks. In Fig. 6(c), where the target exhibits highly similar visual features and temporarily disappears, our method erroneously segments a newly appearing rider upon the target’s return. We attribute this failure to interference caused by the high visual similarity between the original and newly appearing targets after the original object was lost. In Fig. 6(d), our method fails to segment the reappearing adult horse after a long-term disappearance. We believe this is due to the prolonged absence of the target, which overwrote the memory features maintained in the temporal buffer, thereby hindering effective re-identification and segmentation upon reappearance. We plan to address these limitations in future work.

H. Inference Speed and Model Parameters

In our SG-FSCFormer, the trainable modules include PTGFormer, and the MLDecoder, with their respective parameters and FLOPs detailed in Table X. Based on SAM2 and Vicuna-7B, our approach only adds PTGFormer (10.2M) and MLDecoder (85.7M), and achieves 4.96 FPS tested on an A6000 GPU. Furthermore, we compare the inference speed of our method with OVFormer, SAM2, MA-LMM, SMOTer, and VideoGLaMM, which achieve 2.98, 10.61, 7.35, 4.82,

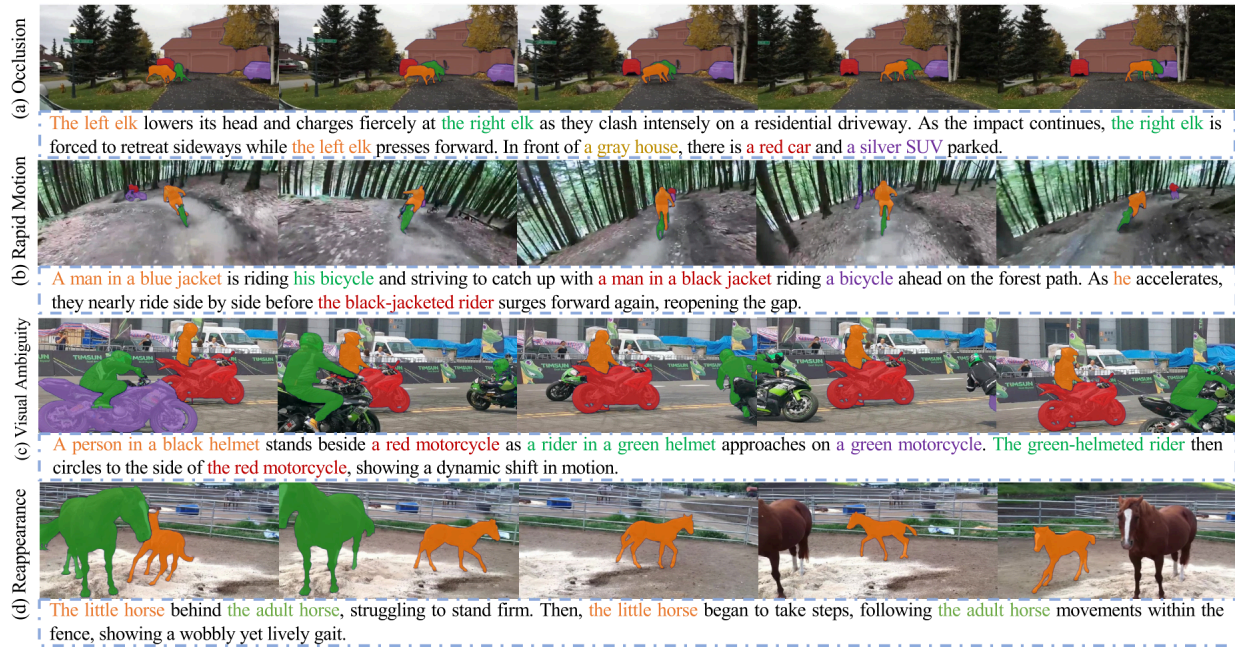


Fig. 6. Visualization of several challenging and failure cases, including heavy occlusion, motion blur, visual ambiguity, and long-term target disappearance.

and 5.13 FPS, respectively, as shown in Table XI. While the inference speed of our model is lower than unimodal generation approaches such as SAM2 for segmentation and MA-LMM for captioning, it outperforms SMOTer, which jointly generates bounding boxes and captions, and achieves inference speed comparable to VideoGLaMM. Overall, our approach introduces relatively few learnable parameters and delivers efficient inference, achieving a favorable balance between model complexity and effectiveness.

I. Limitations

Current video segmentation and captioning methods generate unimodal outputs, limiting the user's ability to access rich multimodal data. Additionally, the integration of video segmentation and captioning often results in oversimplification or omission of content that may be of interest to the user. In contrast, our SegCaptioning task leverages user prompts as guidance and incorporates temporal scene graph modeling, which effectively captures the contextual content of interest and collaboratively generates both video segmentation and contextual descriptions. However, our method relies on manually annotated alignment of masks and captions in two existing video datasets. Scaling this method to larger datasets (e.g., at an Internet scale) remains challenging due to the scarcity of annotated data and high labeling costs. Future work will explore human-supervised semi-automated annotation pipelines and weakly supervised pretraining to overcome this limitation. Additionally, as illustrated in Fig. 6(c) and (d), our method may struggle with targets that exhibit high visual similarity or remain absent from the scene for extended periods. We plan to mitigate these issues by optimizing the memory features maintained in the temporal buffer, enabling more robust long-range modeling and improved target discrimination.

VI. CONCLUSION

This paper introduces a novel research task, "Controllable Video Segmentation and Captioning", establishing it as the inaugural controllable video multimodal interpretation challenge and offering valuable insights to inform future research endeavors. To tackle this task, we propose an innovative framework, "Scene Graph-guided Fine-grained SegCaptioning Transformer", which incorporates a PTGFormer to effectively translate a simple user prompt into prompt-specific graphs, meticulously aligned with user intent. Furthermore, the proposed Mask-linguistic Decoder explicitly enforces alignment between each mask and its corresponding caption tokens, implementing fine-grained alignment to generate precise multimodal outputs that enhance user comprehension of video content. Extensive empirical evaluations on two benchmark datasets substantiate the efficacy of our method, demonstrating its capacity to accurately capture user intent and generate robust multimodal outputs tailored to user needs.

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Xu Zhang received the M.S. degree from Nanjing University of Information Science and Technology, Nanjing, China, in 2019. He is currently pursuing the Ph.D. degree with the College of Computer Science and Electronic Engineering, Hunan University, Changsha, China.

His research interests include computer vision and multimodal scene understanding.



Jin Yuan received the bachelor's and master's degrees from Fudan University in 2005 and 2008, respectively, and the Ph.D. degree in computer science from the National University of Singapore in 2013.

He is currently an Associate Professor with the College of Computer Science and Electronic Engineering, Hunan University. His current research interests include multimedia retrieval and machine learning.



BinHong Yang received the bachelor's degree in computer science and technology from Hainan University in 2023. She is currently pursuing the master's degree with Hunan University.

Her current research interests include machine learning and video moment retrieval.



Internet of Things, and mobile computing.

Xuan Liu (Member, IEEE) received the B.Sc. degree in information and computing mathematics from Xiangtan University, Xiangtan, China, in 2005, the M.Sc. degree in computer science from the National University of Defense Technology, Changsha, China, in 2008, and the Ph.D. degree from The Hong Kong Polytechnic University, Hong Kong, in 2015. She is currently a Professor with the College of Computer Science and Electronic Engineering, Hunan University, Changsha. Her current research interests include machine learning, the



Qianjun Zhang received the Ph.D. degree from Sichuan University in 2017.

She is currently a Postdoctoral Researcher with Southwest Jiaotong University. Her research interests include artificial intelligence, multimodal processing, and large-scale models.



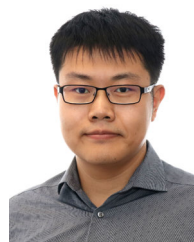
Yuyi Wang received the joint Ph.D. degree from ETH Zurich, Zürich, Switzerland, and KU Leuven, Leuven, Belgium.

He is currently a Youth Scientist with China Railway Rolling Stock Corporation (CRRC) Zhuzhou Institute, leading a Research Laboratory. His research interests include algorithmic theory and artificial intelligence.



Zhiyong Li (Member, IEEE) received the M.Sc. degree in system engineering from the National University of Defense Technology (NUTD), Changsha, China, in 1996, and the Ph.D. degree in control theory and control engineering from Hunan University (HNU), Changsha, in 2004.

In 2004, he joined the College of Computer Science and Electronic Engineering, Hunan University, where he is currently a Full Professor. He has published more than 100 papers in international journals and conferences. His research interests include intelligent perception and autonomous moving body, machine learning and industrial big data, and intelligent optimization algorithms with applications. He is a member of China Computer Federation (CCF) and Chinese Association for Artificial Intelligence (CAAI).



Hanwang Zhang (Senior Member, IEEE) received the B.Eng. degree (Hons.) in computer science from Zhejiang University, Hangzhou, China, in 2009, and the Ph.D. degree in computer science from the National University of Singapore in 2014. He was a Research Scientist with the Department of Computer Science, Columbia University, USA. He is currently an Associate Professor with Nanyang Technological University, Singapore. His research interests include computer vision, multimedia, and social media. He was a recipient of the Best Demo Runner-Up Award

in ACM MM 2012, the Best Student Paper award in ACM MM 2013, the Best Paper Honorable Mention in ACM SIGIR 2016, and the TOMM Best Paper Award in 2018. He is also the winner of the Best Ph.D. Thesis Award of the School of Computing, National University of Singapore, in 2014.